

PHYSICS, CHEMISTRY & MATHEMATICS

PT - I

13.08.2023

Paper – 2

Time Allotted: 3 Hours

Maximum Marks: 204

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

Caution: Question Paper CODE as given above MUST be correctly marked in the answer OMR sheet before attempting the paper. Wrong CODE or no CODE will give wrong results.

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Sections.
3. **Section-I** is Physics, **Section -II** is Chemistry and **Section -III** is Mathematics.
4. Each Section is further divided into two Parts: **Part – A** & **Part – C** based on type of questions as this paper doesn't have matrix matching so there is no **Part B** in question paper.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with a **blue / black ball point pen** for each character of your Enrolment No. and write in ink your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Sections.

- Part-A (1 – 6)** contains 6 multiple choice questions which have **only one** correct answer. Each question carries **+3 marks** for correct answer and **–1 mark** for wrong answer.

Part-A (7 – 14) contains 8 multiple choice questions which have **one or more than one** correct answer(s). Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.
- Part-C (1 – 6)** contains 6 Numerical based questions with single digit integer as answer, ranging from 0 to 9 (both inclusive) and each question carries **+3 marks** for correct answer and **–1 mark** for wrong answer.

Name of the Candidate : _____

Batch : _____ Date of Examination : _____

Enrolment Number : _____

Batch: LKTYR235A03

USEFUL DATA

PHYSICS

Acceleration due to gravity	: $g = 10 \text{ m/s}^2$
Planck constant	: $h = 6.6 \times 10^{-34} \text{ J-s}$
Charge of electron	: $e = 1.6 \times 10^{-19} \text{ C}$
Mass of electron	: $m_e = 9.1 \times 10^{-31} \text{ kg}$
Permittivity of free space	: $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2 / \text{N-m}^2$
Density of water	: $\rho_{\text{water}} = 10^3 \text{ kg/m}^3$
Atmospheric pressure	: $P_a = 10^5 \text{ N/m}^2$
Gas constant	: $R = 8.314 \text{ J}$

 $K^{-1} \text{ mol}^{-1}$

CHEMISTRY

Gas Constant	R =	$8.314 \text{ J K}^{-1} \text{ mol}^{-1}$
	=	0.0821 Lit atm
$K^{-1} \text{ mol}^{-1}$		
	=	1.987 $\approx 2 \text{ Cal}$
$K^{-1} \text{ mol}^{-1}$		
Avogadro's Number N_a	=	6.023×10^{23}
Planck's constant h	=	$6.625 \times 10^{-34} \text{ J.s}$
	=	$6.625 \times 10^{-27} \text{ erg.s}$
1 Faraday	=	96500 coulomb
1 calorie	=	4.2 joule
1 amu	=	$1.66 \times 10^{-27} \text{ kg}$
1 eV	=	$1.6 \times 10^{-19} \text{ J}$

Atomic No: H = 1, He = 2, Li = 3, Be = 4, B = 5, C = 6, N = 7, O = 8, F = 9, Ne = 10, Na = 11, Mg = 12, Si = 14, Al = 13, P = 15, S = 16, Cl = 17, Ar = 18, K = 19, Ca = 20, Cr = 24, Mn = 25, Fe = 26, Co = 27, Ni = 28, Cu = 29, Zn = 30, As = 33, Br = 35, Ag = 47, Sn = 50, I = 53, Xe = 54, Ba = 56, Pb = 82, U = 92.

Atomic masses: H = 1, He = 4, Li = 7, Be = 9, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Si = 28, Al = 27, P = 31, S = 32, Cl = 35.5, K = 39, Ca = 40, Cr = 52, Mn = 55, Fe = 56, Co = 59, Ni = 58.7, Cu = 63.5, Zn = 65.4, As = 75, Br = 80, Ag = 108, Sn = 118.7, I = 127, Xe = 131, Ba = 137, Pb = 207, U = 238.

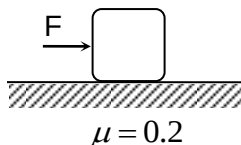
Section – I (Physics)

PART – A

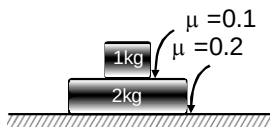
(Single Correct Choice Type)

This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

1. A body of mass 2 kg is placed on rough horizontal plane. The coefficient of friction between body and plane is 0.2. Then:

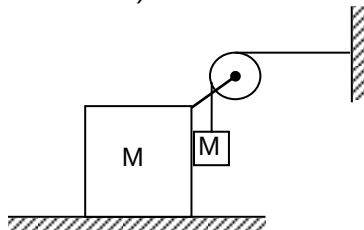


- (A) Body will move in forward direction if $F = 5 \text{ N}$
 (B) Body will move in backward direction with acceleration 0.5 m/s^2 if force $F = 3 \text{ N}$
 (C) If $F = 3 \text{ N}$, then body will be in rest condition
 (D) Both (A) and (C) are correct
2. Both the blocks shown in the given arrangement are given together a horizontal velocity towards right. If a_{CM} be the subsequent acceleration of the centre of mass of the system of blocks, then a_{CM} equals :



- (A) 0 m/s^2
 (B) $5/3 \text{ m/s}^2$
 (C) $7/3 \text{ m/s}^2$
 (D) 2 m/s^2
3. A stone is dropped from a balloon ascending with velocity 2 m/s from a height of 20 m . The time of flight of the stone is
 (A) 1 sec.
 (B) 2 sec.
 (C) 4 sec.
 (D) data insufficient
4. The equation of trajectory of a particle is given by the equation $y = ax - bx^2$, where a and b are constant. The horizontal range is
 (A) a
 (B) b
 (C) ab
 (D) a/b
5. When forces F_1 , F_2 and F_3 are acting on a particle of mass m such that F_2 and F_3 are mutually perpendicular, then the particle remains stationary. If the force F_1 is now removed, then the acceleration of the particle is
 (A) $\frac{F_1}{m}$
 (B) $\frac{F_2 F_3}{m F_1}$
 (C) $\frac{(F_2 - F_3)}{m}$
 (D) $\frac{F_2}{m}$

6. In the given arrangement value N for both the blocks is (assume pulleys to be ideal)



- (A) $2M$ (B) $\frac{Mg}{2}$
(C) $\frac{3}{2}M$ (D) M

PART – A

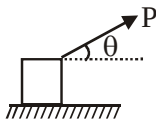
(Multi Correct Choice Type)

This section contains **8 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.

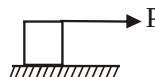
7. Figure shows three blocks on a rough surface under influence of a force P of same magnitude in all the three cases. Coefficient of friction is same between each block and ground. What possible relation holds between magnitudes of normal reaction and friction forces. (Assume that blocks do not overturn about edge.) Here, f_A , f_B & f_C are frictional forces & N_A , N_B & N_C are normal reactions.



(A)



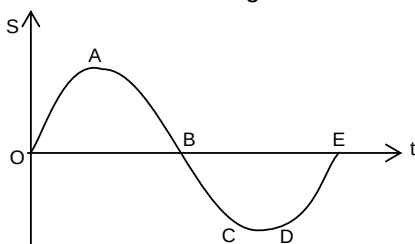
(B)



(C)

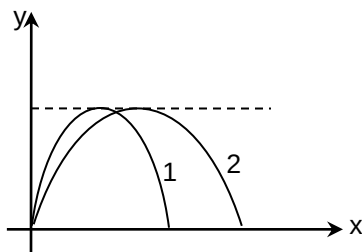
- (A) $N_A > N_C > N_B$ (B) $f_A > f_C > f_B$
(C) $f_C > f_A = f_B$ (D) $N_C > N_A = N_B$

8. A particle has a rectilinear motion and the figure gives its displacement as a function of time. Which of the following statements are true with respect to the motion.

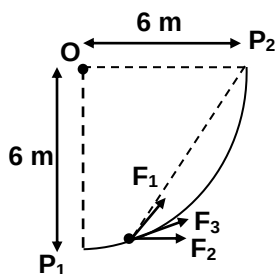


- (A) in motion between 0 to A, the velocity is positive and acceleration is negative
(B) between A and B, the velocity and acceleration are positive
(C) between B and C, the velocity is negative and acceleration is positive
(D) between D and E, the acceleration is positive

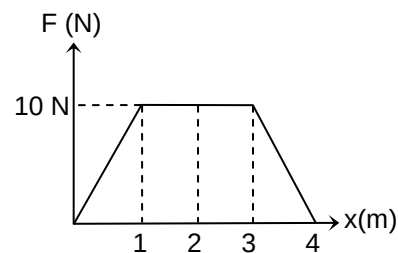
9. Trajectories of two projectiles are shown in figure. Let T_1 and T_2 be the time periods and u_1 and u_2 their speeds of projection. Then



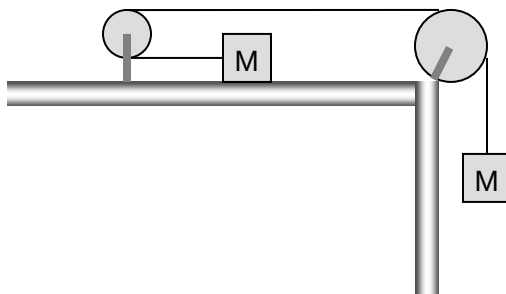
- (A) $T_2 > T_1$ (B) $T_1 = T_2$
 (C) $u_1 > u_2$ (D) $u_1 < u_2$
10. A smooth track in form of a quarter circle of radius 6 m lies in the vertical plane. A particle moves from P_1 to P_2 under the action of forces $\vec{F}_1, \vec{F}_2$ and $\vec{F}_3$ force $\vec{F}_1$ is always towards P_2 and is always 20 N in magnitude force $\vec{F}_2$ always acts horizontally and is always 30 N in magnitude. Force $\vec{F}_3$ always acts tangentially to the track and is of magnitude 15 N. Select the correct statements



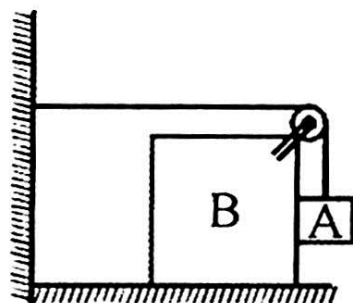
- (A) work done by $\vec{F}_1$ is $120\sqrt{2}$ J (B) work done by $\vec{F}_2$ is 180 J
 (C) work done by $\vec{F}_3$ is 95 J (D) $\vec{F}_1$ is conservative in nature
11. Figure shows the force F (in newton) acting on a body as a function of x (displacement), then select correct options. (Assuming mass is 1 kg, velocity is zero at origin).
- (A) At $x = 1$ m, velocity is $\sqrt{10}$ m/s
 (B) At $x = 1$ m, velocity is $\sqrt{20}$ m/s
 (C) At $x = 4$ m, velocity is $\sqrt{60}$ m/s
 (D) At $x = 4$ m, velocity is $\sqrt{80}$ m/s



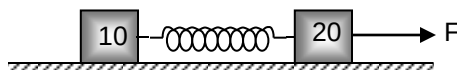
12. Calculate the acceleration of either blocks and tension in the string shown in figure. The pulley and the string are light and all surfaces are smooth.



- (A) $T = Mg/2$
 (B) $T = Mg/4$
 (C) $a = g/2$
 (D) $a = g/4$
13. In the system shown in the diagram all surfaces are smooth; pulley and strings are ideal. If $\vec{a}_A$ and $\vec{a}_B$ are the acceleration of the two blocks, then just after the system is released from rest, then choose incorrect statement(s):



- (A) $|\vec{a}_A| = |\vec{a}_B|$
 (B) $\vec{a}_A \perp \vec{a}_B$
 (C) Acceleration of A relative to B is vertically downwards
 (D) Normal force exerted by B on A is zero
14. Two masses of 10 kg and 20 kg are connected by a light spring as shown. A force of 200 N acts on 20 kg mass. At a certain instant, the acceleration of 10 kg mass is 12 m/s^2 .



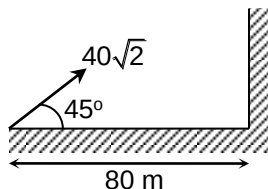
- (A) At that instant 20 kg mass has acceleration 2 m/s^2
 (B) At that instant 20 kg mass has acceleration 4 m/s^2
 (C) The stretching force in the spring is 120 N
 (D) Spring force have different magnitudes for both blocks.

PART – C

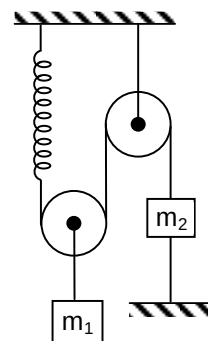
(Numerical Based)

This section contains **6 questions**. The answer to each question is a single digit integer, ranging from 0 to 9 (both inclusive).

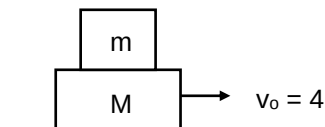
1. Speed of particle when it strikes the wall is v then the value of $v / 4\sqrt{5}$



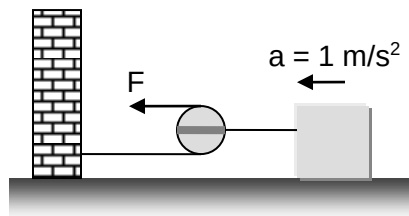
2. In the given figure, pulleys are ideal. Initially the system is in equilibrium and string connecting m_2 to rigid support is cut. Find initial acceleration (in m/s^2) of m_2 (Take $m_1 = 15 \text{ kg}$, $m_2 = 5 \text{ kg}$, $g = 10 \text{ m/s}^2$)



3. A stretching force of 100 N is applied at one end of a spring balance and an equal stretching force is applied at the other end at the same time. If the reading of the balance is $5x \text{ kg}$, find the value of x . ($g = 10 \text{ m/s}^2$)
4. A stationary body of mass m is slowly lowered onto a massive platform of mass M ($M \gg m$) moving at a speed $v_0 = 4 \text{ m/s}$. How far (in meter) will the body slide along the platform ($\mu = 0.2$ and $g = 10 \text{ m/s}^2$)?



5. A ball of mass m is suspended from a massless thread which is fastened at point O. The point of suspension, O is displaced in vertical direction with an acceleration such that the tension in the thread is half the weight of the ball. Find the acceleration of the ball in m/s^2 ($g = 10 \text{ m/s}^2$)
6. A block of mass 200 kg is set into motion on a frictionless horizontal surface with the help of frictionless pulley and a rope system as shown in figure. A horizontal force $F = 50n \text{ N}$ applied to produce an acceleration of 1 m/s^2 in the block. Find n .



Section – II (Chemistry)

PART – A

(Single Correct Choice Type)

This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

1. The correct order of hybridisation for CO_2 , I_3^- , XeF_2
 - (A) all sp
 - (B) sp, sp^2 , sp^3
 - (C) sp, sp^3 , sp^3d
 - (D) sp, sp^3d , sp^3d
2. Which of the atomic orbitals cannot overlap to form a bond?
 - (A) $1s$ and $2p_x$
 - (B) $2p_x$ and $2p_y$
 - (C) $3p_y$ and $3p_z$
 - (D) $1s$ and $1s$
3. The shape and hybridization of XeF_4 is
 - (A) Planar, sp^3
 - (B) Planar, sp^3d^2
 - (C) Octahedral, sp^3d^2
 - (D) Tetrahedral, sp^3
4. Number of radial nodes in $2s$ orbital is
 - (A) 0
 - (B) 1
 - (C) 2
 - (D) 3
5. If the radius of 2^{nd} Bohr orbit of hydrogen atom is r_2 , the radius of third Bohr orbit will be
 - (A) $\frac{4}{9}r_2$
 - (B) $4r_2$
 - (C) $\frac{9}{4}r_2$
 - (D) $9r_2$
6. Number of waves made by an electron in one complete revolution in 3^{rd} Bohr orbit is
 - (A) 2
 - (B) 3
 - (C) 4
 - (D) 1

PART – A

(Multi Correct Choice Type)

This section contains **8 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.

7. Select the correct option(s)
 - (A) O_2 bond order is 2 and it is paramagnetic
 - (B) O_2^+ bond order is 2.5 and it is paramagnetic
 - (C) O_2^- bond order is 1.5 and it is paramagnetic
 - (D) Be_2 does not exist
8. The species(s) having sp^3 hybridization is/are
 - (A) H_2O
 - (B) NH_3
 - (C) NH_4^+
 - (D) NH_2^-
9. Select correct order between following compounds:
 - (A) $\text{NH}_3 > \text{NF}_3$:Bond angle
 - (B) $\text{NH}_3 > \text{NF}_3$:Dipole moment
 - (C) $\text{HF} > \text{H}_2\text{O}$:Boiling point
 - (D) Water > Ice: Density

10. Which of the following statement is/are correct?
 (A) Number of radial nodes for 3d orbital is zero
 (B) Number of angular nodes for 3d orbital is 2
 (C) Number of nodal planes for 2p orbital is 1
 (D) Total number of nodes in 3s orbital is 2
11. The set of quantum numbers which are impossible for an electron are
 (A) $n = 1, \ell = 0, m_\ell = 1, m_s = +\frac{1}{2}$ (B) $n = 2, \ell = 2, m_\ell = 1, m_s = -\frac{1}{2}$
 (C) $n = 1, \ell = 0, m_\ell = 0, m_s = +\frac{1}{2}$ (D) $n = 2, \ell = 1, m_\ell = 2, m_s = -\frac{1}{2}$
12. In halogens, which of the followings increase from iodine to fluorine?
 (A) Bond length (B) Electronegativity
 (C) IE_1 (D) Electron affinity
13. 11.2 L of a gas at STP weight 14 g. The gas could be
 (A) N_2 (B) CO
 (C) NO_2 (D) N_2O
14. Number of atoms in 20 gm Ca is equal to the number of atoms in
 (A) 20 gm Mg (B) 1.6 gm CH_4
 (C) 1.8 gm H_2O (D) 2.125 gm NH_3

PART – C

(Numerical Based)

This section contains **6 questions**. The answer to each question is a single digit integer, ranging from 0 to 9 (both inclusive).

1. How many unpaired electrons in O_2 molecule?
2. How many of the following reactions shows decrease in bond length?
 (A) $CO \longrightarrow CO^+$ (B) $NO \longrightarrow NO^+$
 (C) $CN \longrightarrow CN^-$ (D) $N_2 \longrightarrow N_2^+$
 (e) $O_2 \longrightarrow O_2^+$ (f) $O_2 \longrightarrow O_2^{-1}$
 (g) $O_2 \longrightarrow O_2^{-2}$ (h) $CN \longrightarrow CN^+$
3. The wavelength of a Balmer line of He^+ ion is given as $\frac{4}{3R}$. What is the principal quantum number of the higher orbit, from which the electron transition has taken place?
 [R = Rydberg constant]
4. The maximum number of electrons that can have principal quantum number $n = 3$ and spin quantum number $m_s = -\frac{1}{2}$ is
5. The hydrated salt $Na_2SO_4 \cdot nH_2O$ undergoes 50.3% loss in weight on heating and become anhydrous. The value of 'n' will be _____.
6. Caffeine has a molecular weight of 194. If it contains 28.9% by mass of Nitrogen, number of Nitrogen in one molecule of Caffeine is?

Section – III (Mathematics)

PART – A

(Single Correct Choice Type)

This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

1. $3(\sin x - \cos x)^4 + 6(\sin x + \cos x)^2 + 4(\sin^6 x + \cos^6 x) =$
 (A) 11 (B) 12
 (C) 13 (D) 14
2. Which of the following is a rational number?
 (A) $\sin 15^\circ$ (B) $\cos 15^\circ$
 (C) $\sin 15^\circ \cdot \cos 15^\circ$ (D) $\sin 15^\circ \cdot \cos 75^\circ$
3. The equation to the pair of straight lines through the origin and perpendicular to $2x^2 - 4xy - 7y^2 = 0$ is
 (A) $7x^2 - 4xy - 2y^2 = 0$ (B) $7x^2 + 4xy - 2y^2 = 0$
 (C) $7x^2 + 4xy + 2y^2 = 0$ (D) $-7x^2 + 4xy - 2y^2 = 0$
4. The parametric equation of a line is given by $x = -2 + \frac{r}{\sqrt{10}}$ & $y = 1 + 3\frac{r}{\sqrt{10}}$. Then, for the line
 (A) intercept on the x-axis = $\frac{7}{3}$ (B) intercept on the y-axis = -7
 (C) slope of the line = $\tan^{-1} \frac{1}{3}$ (D) slope of the line = $\tan^{-1} 3$
5. P & Q are any two points on the circle $x^2 + y^2 = 4$ such that PQ is a diameter. If α and β are length of perpendicular from P and Q on $x + y = 1$, then the maximum value of $\alpha\beta$ is
 (A) $\frac{1}{2}$ (B) $\frac{7}{2}$
 (C) 1 (D) 2
6. From a point R(5,8) two tangents RP and RQ are drawn to a given circle $S=0$ whose radius is 5. If circumcentre of the triangle PQR is (2,3), then the equation of circle $S=0$ is
 (A) $x^2 + y^2 + 2x + 4y - 20 = 0$ (B) $x^2 + y^2 + x + 2y - 10 = 0$
 (C) $x^2 + y^2 - x - 2y - 20 = 0$ (D) $x^2 + y^2 - 4x - 6y - 12 = 0$

PART – A

(Multi Correct Choice Type)

This section contains **8 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE OR MORE** may be correct.

7. The equation of lines passing through point (2, 3) and having an intercept of length 2 units between the lines $2x + y = 3$ and $2x + y = 5$ are
 (A) $y = 3$ (B) $x = 2$ (C) $y = x + 1$ (D) $4y + 3x = 18$

8. If β, γ be two positive angles $< \pi$ and $\cos \alpha + \cos(\alpha + \beta) + \cos(\alpha + \beta + \gamma) = 0$, and $\sin \alpha + \sin(\alpha + \beta) + \sin(\alpha + \beta + \gamma) = 0$, then
- (A) $\beta + \gamma = \frac{4\pi}{3}$ (B) $\beta + \gamma = \frac{2\pi}{3}$
 (C) $\beta - \gamma = 0$ (D) $\beta - \gamma = \frac{\pi}{3}$
9. The locus of points of intersection of the tangents to $x^2 + y^2 = a^2$ at the extremities of a chord of circle $x^2 + y^2 = a^2$ which touches the circle $x^2 + y^2 - 2ax = 0$ is/are :
- (A) $y^2 = a(a - 2x)$ (B) $y^2 = a(a - 2y)$
 (C) $x^2 + y^2 = (x - a)^2$ (D) $x^2 + y^2 = (y - a)^2$
10. Let $f_n(\theta) = \frac{\cos \frac{\theta}{2} + \cos 2\theta + \cos \frac{7\theta}{2} + \dots + \cos(3n-2)\frac{\theta}{2}}{\sin \frac{\theta}{2} + \sin 2\theta + \sin \frac{7\theta}{2} + \dots + \sin(3n-2)\frac{\theta}{2}}$ then
- (A) $f_1\left(\frac{\pi}{2}\right) = 1$ (B) $f_2\left(\frac{\pi}{2}\right) = 1 - \sqrt{2}$
 (C) $f_1(\pi) = 0$ (D) $f_1(\pi) = 1$
11. Equation of the straight line passing through the point (4,5) and equally inclined to the lines $3x = 4y + 7$ and $5y = 12x + 6$ is
- (A) $9x - 7y = 1$ (B) $9x + 7y = 71$
 (C) $7x + 9y = 73$ (D) $7x - 9y + 17 = 0$
12. The points $A(0,0)$, $B(\cos \alpha, \sin \alpha)$, and $C(\cos \beta, \sin \beta)$ are the vertices of a right-angled triangle if
- (A) $\sin \frac{\alpha - \beta}{2} = \frac{1}{\sqrt{2}}$ (B) $\cos \frac{\alpha - \beta}{2} = -\frac{1}{\sqrt{2}}$
 (C) $\cos \frac{\alpha - \beta}{2} = \frac{1}{\sqrt{2}}$ (D) $\sin \frac{\alpha - \beta}{2} = -\frac{1}{\sqrt{2}}$
13. If one vertex of an equilateral triangle of side 2 is the origin and another vertex lies on the line $x = \sqrt{3}y$ then the third vertex can be
- (A) (0, 2) (B) $(\sqrt{3}, -1)$
 (C) (0, -2) (D) $(\sqrt{3}, 1)$
14. If the equations of the three sides of a triangle are $x + y = 1$, $3x + 5y = 2$ and $x - y = 0$ then the orthocentre of the triangle lies on the line
- (A) $5x - 3y = 2$ (B) $3x - 5y + 1 = 0$
 (C) $2x - 3y = 1$ (D) $5x - 3y = 1$

PART – C**(Numerical Based)**

This section contains **6 questions**. The answer to each question is a single digit integer, ranging from 0 to 9 (both inclusive).

- Two consecutive side of a parallelogram are $x - 2y + 2 = 0$ and $2x + y - 16 = 0$. One of the diagonals of the parallelogram is $3x - y - 9 = 0$ then area of parallelogram is (in sq. units)
- Sum of all the integral solution of the inequality $|x - 1| + |x - 2| \leq 3$ is equal to _____
- If $\sin 32^\circ = k$ and $\cos x = 1 - 2k^2$; α, β are the values of x between 0° and 360° with $\alpha < \beta$, then the value in degrees of $\frac{\beta - 4\alpha}{10}$ is equal to
- A circle of unit radius touches positive x -axis and y -axis at A and B, respectively. A variable line passing through origin intersects the circle in two points D and E. If the slope of the line is m and the area of $\triangle DEB$ is maximum, then the value of $\frac{1}{m^2}$ must be
- Two circle are inscribed and circumscribed about a square ABCD, length of each side of the square is 16. P and Q are two points respectively on these circles, then $\frac{\sum(PA)^2 + \sum(QA)^2}{256}$ is equal to.....
- The number of integral values of a for which the point $(a - 1, a + 1)$ lies in the larger segment of the circles $x^2 + y^2 - x - y - 6 = 0$ made by chord whose equation is $x + y - 2 = 0$ is

FIITJEE INTERNAL TEST

Batch: LKTYR235A03

IIT - JEE 2025

PHASE – I PAPER – II

ANSWERS

QP CODE:

Physics

Part – A

- | | | | |
|---------|---------|--------|--------|
| 1. D | 2. D | 3. B | 4. D |
| 5. A | 6. B | 7. ABC | 8. ACD |
| 9. BD | 10. ABD | 11. AC | 12. AC |
| 13. ABD | 14. BC | | |

Part – C

- | | | | |
|------|------|------|------|
| 1. 5 | 2. 5 | 3. 2 | 4. 4 |
| 5. 5 | 6. 2 | | |

Chemistry

Part – A

- | | | | |
|--------|----------|---------|---------|
| 1. D | 2. B | 3. B | 4. B |
| 5. C | 6. B | 7. ABCD | 8. ABCD |
| 9. ABD | 10. ABCD | 11. ABD | 12. BC |
| 13. AB | 14. BD | | |

Part – C

- | | | | |
|------|------|------|------|
| 1. 2 | 2. 4 | 3. 4 | 4. 9 |
| 5. 8 | 6. 4 | | |

Mathematics

Part – A

- | | | | |
|---------|---------|--------|----------|
| 1. C | 2. C | 3. A | 4. D |
| 5. B | 6. A | 7. BD | 8. AC |
| 9. AC | 10. ABC | 11. AC | 12. ABCD |
| 13. ABC | 14. BD | | |

Part – C

- | | | | |
|------|------|------|------|
| 1. 5 | 2. 6 | 3. 4 | 4. 3 |
| 5. 7 | 6. 1 | | |